



# Modeling direct air carbon capture and storage in a 1.5 °C climate future using historical analogs

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Limiting the rise in global temperature to 1.5 °C will rely, in part, on technologies to remove CO<sub>2</sub> from the atmosphere. However, many carbon dioxide removal (CDR) technologies are in the early stages of development, and there is limited data to inform predictions of their future adoption. Here, we present an approach to model adoption of early-stage technologies such as CDR and apply it to direct air carbon capture and storage (DACCS). Our approach combines empirical data on historical technology analogs and early adoption indicators to model a range of feasible growth pathways. We use these pathways as inputs to an integrated assessment model (the Global Change Analysis Model, GCAM) and evaluate their effects under an emissions policy to limit end-of-century temperature change to 1.5 °C. Adoption varies widely across analogs, which share different strategic similarities with DACCS. If DACCS growth mirrors high-growth analogs (e.g., solar photovoltaics), it can reach up to 4.9 GtCO<sub>2</sub> removal by midcentury, compared to as low as 0.2 GtCO<sub>2</sub> for low-growth analogs (e.g., natural gas pipelines). For these slower growing analogs, unabated fossil fuel generation in 2050 is reduced by 44% compared to high-growth analogs, with implications for energy investments and stranded assets. Residual emissions at the end of the century are also substantially lower (by up to 43% and 34% in transportation and industry) under lower DACCS scenarios. The large variation in growth rates observed for different analogs can also point to policy takeaways for enabling DACCS.

modeling for sustainability | carbon dioxide removal | innovation systems | net zero pathways | direct air carbon capture and storage

Since the Paris Agreement in 2015, a wave of actors have set targets to reach net zero carbon dioxide (CO<sub>2</sub>) emissions. Emissions pathways consistent with a 1.5 °C temperature target reach net zero by approximately 2050 (or 2070 for a 2 °C target) (1). Since some emissions sources are difficult to eliminate, and because overshooting targets is likely, reaching net zero will almost certainly require technologies to remove CO<sub>2</sub> from the atmosphere alongside deep emissions reductions (2). Integrated assessment model (IAM) scenarios targeting 1.5 °C frequently feature gigatons (Gt) of carbon dioxide removal (CDR) per year, on the order of 25% or more of current emissions (1). CDR involves capturing ambient CO<sub>2</sub> and sequestering it underground, on land, in the sea, or in long-lived products. Despite the major role it plays in mitigation scenarios, the future prospects for CDR are highly uncertain. There is thus an urgent need to understand the potential and limits of CDR for climate policy.

A variety of approaches can be used to capture and sequester CO<sub>2</sub> from the atmosphere (3). Some approaches, such as afforestation and reforestation, have been used for a long time and represent virtually all current CDR (4). Other approaches, such as bioenergy with carbon capture and storage (BECCS) and direct air carbon capture and storage (DACCS), are in the early stages of development and make up a tiny fraction of current removals. BECCS converts biomass to useful energy and captures the resulting CO<sub>2</sub> emissions. DACCS removes CO<sub>2</sub> directly from ambient air (5). With either technology, captured CO<sub>2</sub> is compressed, transported, and permanently stored underground or repurposed for industrial uses, building materials, and other products. Historically, BECCS (along with afforestation) was the major CDR technology modeled in IAM scenarios. However, research and policy interest in DACCS has grown substantially in recent years as technology innovation by start-ups and demand pull through corporate and public purchase agreements are accelerating early adoption (6, 7).

Researchers and practitioners have questioned society's ability to reach Gt-scale CDR from novel approaches such as BECCS and DACCS, given the small role these technologies play in climate change mitigation today (8, 9). Others have highlighted the potential environmental (10–12) and social [e.g., food prices (11)] impacts of CDR, particularly

## Significance

Technological innovation is central to sustainable development, but representing novel technologies in systems models is difficult due to limited data on their past performance. We propose a method to model the feasibility space for novel technologies that combines empirical data on historical analogs and early adoption with a global integrated assessment model. Applying this method to direct air carbon capture and storage (DACCS), we find that the feasibility space is large, with DACCS contributing meaningfully to net-zero goals if it grows like some analogs and failing to do so with others. The results can be used to identify technology and policy features that may be important in enabling rapid adoption to avert the worst effects of climate change.

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for BECCS due to its high land and water requirements but also for DACCS. They have also critiqued the role that CDR plays in net-zero policy narratives, arguing that optimistic assumptions about CDR in the future may be used to delay action today and represent a moral hazard whose risks are disproportionately borne by low-income countries and future generations (13, 14). While research on institutional, behavioral, and social barriers and enablers for CDR is more limited (15), experience with related technologies suggests that they may be substantial (16, 17). However, despite growing interest in CDR, there is a lack of understanding of potential future growth pathways, which features of technologies enable or constrain these pathways, and how they can be represented in systems models to inform climate policy.

The critiques of CDR in IAMs are emblematic of a broader challenge: Innovation and novelty are defining features in nature-society systems (18), yet they can be difficult to model. There is a large empirical and theoretical literature on technological innovation that can inform the design and evaluation of models (refs. 19–21 and *SI Appendix, Note S1*). Previous work merging insights from innovation systems and systems modeling has argued that growth rates observed in IAMs are too fast or too slow (22) (see *SI Appendix, Note S5* for an overview of the feasibility literature). Others take a technology-specific approach and argue that IAMs are biased toward large, infrastructure-intensive climate solutions (23–25). Despite these critiques, IAMs are widely used to study the effects of innovation on the costs of climate policy and the future portfolio of low-carbon technologies (26). Recent advances in model development have targeted limitations of IAMs for these applications by improving the representations of technological change, growth rates, and diffusion processes (27).

This paper aims to bridge theory and methods from innovation systems and integrated assessment modeling to advance approaches to modeling the adoption feasibility space for novel technologies such as CDR. We combine data on historical technology analogs and early adoption indicators to design scenarios in a state-of-the-art global IAM. We advance prior work in this area in three key ways. First, we use a systematic approach to identify multiple analogs for DACCS, with a focus on strategic similarity across multiple social and technical dimensions. Second, we use these analogs to characterize uncertainty in future technology growth. Third, we use an IAM to explore the effects of different DACCS growth pathways on sociotechnical transitions under a climate policy. Our approach complements recent advances that constrain the rate of technology growth (28, 29) or retirement (30) in systems models, use historical analogs to project adoption outside systems models (31, 32), or assess the feasibility of mitigation scenarios (33, 34). Specifically, we address the question: What can we learn from technologies from the past about the role of DACCS in climate policy?

## Results

Our research approach consists of multiple steps: 1) select historical technology analogs for the target innovation (i.e., DACCS), 2) estimate early adoption using data on start-up capacity announcements, 3) model a set of potential growth pathways using different historical analogs, and 4) input these pathways in an integrated assessment model to generate insights on their system-wide effects. We describe the results of each of these steps in the four sections that follow. Further details on our methodological approach are provided in the *Materials and Methods*.

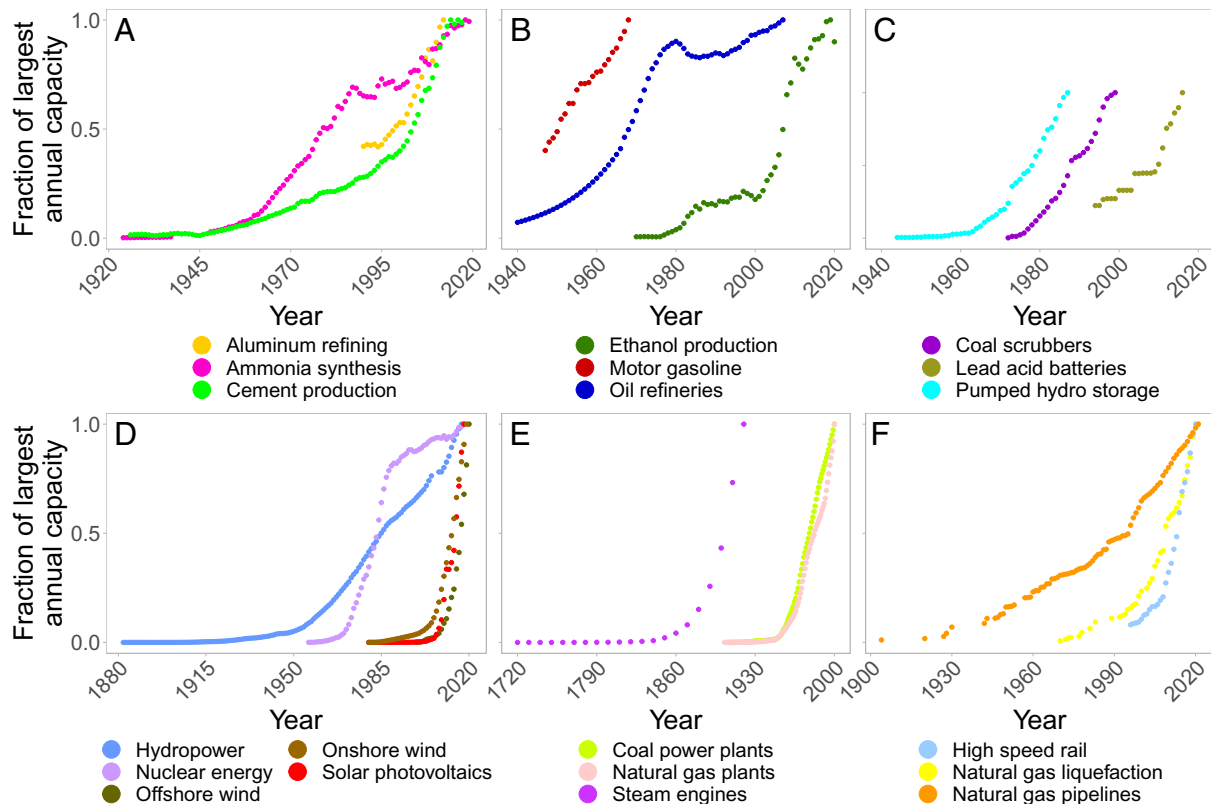
**Historical Technology Analogs.** Researchers have identified many factors that partially explain the large variation in growth rates observed for historical technologies over time, from the formative

phase (which can be irregular and highly uncertain) to the growth phase and eventually to market saturation (35, 36). Formative phases are typically slow for energy technologies, averaging 22 y but potentially ranging from 4 to 168 y (*SI Appendix, Note S2*) (37). Across these phases, faster cost reductions and higher growth rates are associated with smaller technologies that have simple designs, modular construction, high possibility of mass manufacturing, and low investment risk (38–40). Slow expansion and modest cost reductions (and in some cases cost increases) are linked to large, custom-built, capital-intensive, “lumpy” technologies, or those with high complexity, infrastructure needs, coordination requirements, and investment risks (38–41). These features are thus important for identifying appropriate analogs for DACCS and for tailoring public policies to accelerate innovation (39).

DACCS is an emerging technology with multiple promising designs. These designs fall into three categories: solid (S-DAC), liquid (L-DAC), and other, which include a range of less-developed designs (e.g., metal-organic framework materials). S-DAC is a modular technology that uses solid sorbent materials to adsorb CO<sub>2</sub> and then varies pressure or humidity to release the CO<sub>2</sub> in a purified stream (5). S-DAC operates at relatively low temperatures and can run on a heat pump (42, 43). L-DAC is a two-step process that relies on strong aqueous alkaline solutions to absorb CO<sub>2</sub> and produce pelletized calcium carbonate (5, 44). High temperatures, typically achieved by burning natural gas, are then used to produce a purified stream of CO<sub>2</sub> and regenerate the solvents (42, 44). While current S-DAC systems have lower life cycle emissions, L-DAC benefits from efficiencies of scale and may perform favorably at larger unit scales (43).

We can gain insight into future DACCS adoption by looking at similar technologies from the past. To do this, we apply the systematic historical analog research for decision-making (SHARD) method, in which a target innovation is matched with historical analogs according to sociotechnical characteristics from the sustainability transitions literature (see *SI Appendix, Note S4* for background on SHARD and *Materials and Methods* and *SI Appendix, section B* for full details on our approach) (45). We identify a set of analogs that share strategic similarities with DACCS across six categories (Fig. 1): chemicals and materials, fuel production, enabling technologies, low-carbon generation, fossil fuel combustion, and transportation/infrastructure. We also include a comparison with solar photovoltaics (PV), which is technologically less similar to DACCS but frequently used as a fast-growth bounding case (31, 32, 46). Among these analogs, the range of feasible growth rates for DACCS varies by an order of magnitude, from 3.2% for gas pipelines to 29% for offshore wind (and 31% for solar PV).

**Early Adoption Indicators.** Public and private sector investment in new direct air capture (DAC) capacity has grown rapidly in recent years, both with permanent CO<sub>2</sub> storage (referred to as DACCS) and without. Start-ups have a variety of plans for the fate of the CO<sub>2</sub> that will be captured, including geologic sequestration, utilization in products (e.g., Global Thermostat selling CO<sub>2</sub> to soda bottlers), and enhanced oil recovery (EOR) (e.g., 1PointFive’s partnership with Occidental Petroleum). While EOR technically does permanently store captured CO<sub>2</sub> and may provide an early application where DAC technologies can develop and improve, supporting DAC start-ups using captured CO<sub>2</sub> for EOR can incentivize the prolonged use of fossil fuels and ultimately be counterproductive for climate policy. [The US Department of Energy requires that CO<sub>2</sub> utilization be included in the system boundaries for plants applying for funding (47).] While we do not take a position on such support here, we use empirical data on all DAC (with or without storage) given its



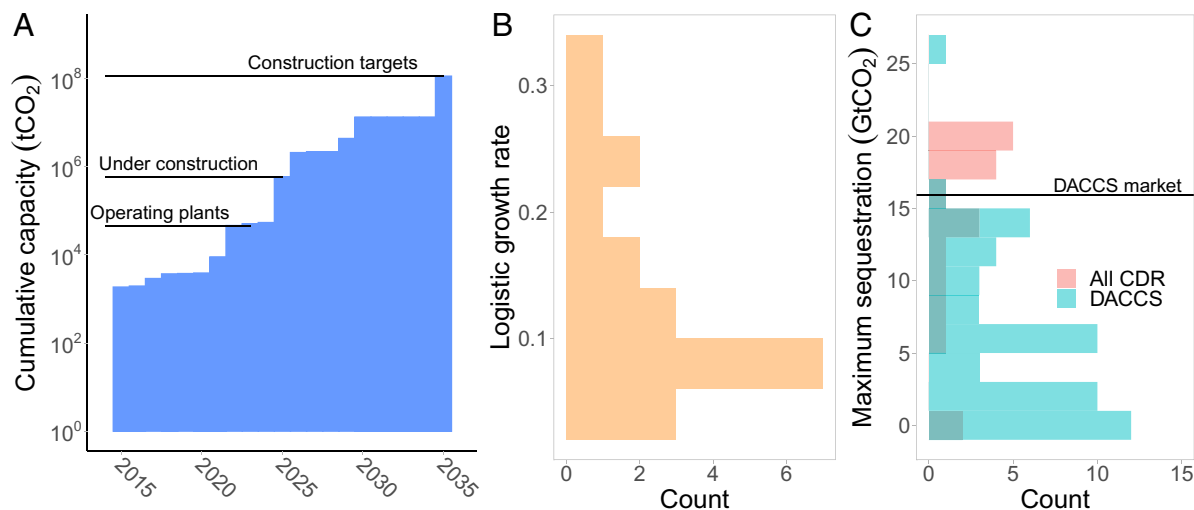
**Fig. 1.** Historical technology analogs for DACCS presented in six groups based on technology category: (A) chemicals and materials, (B) fuel production, (C) enabling technologies, (D) low-carbon generation, (E) fossil fuel combustion, and (F) transport/infrastructure. Data are sourced from HATCH 1.0 (*Materials and Methods*).

role as an indicator of advancing innovation, market creation, and future growth.

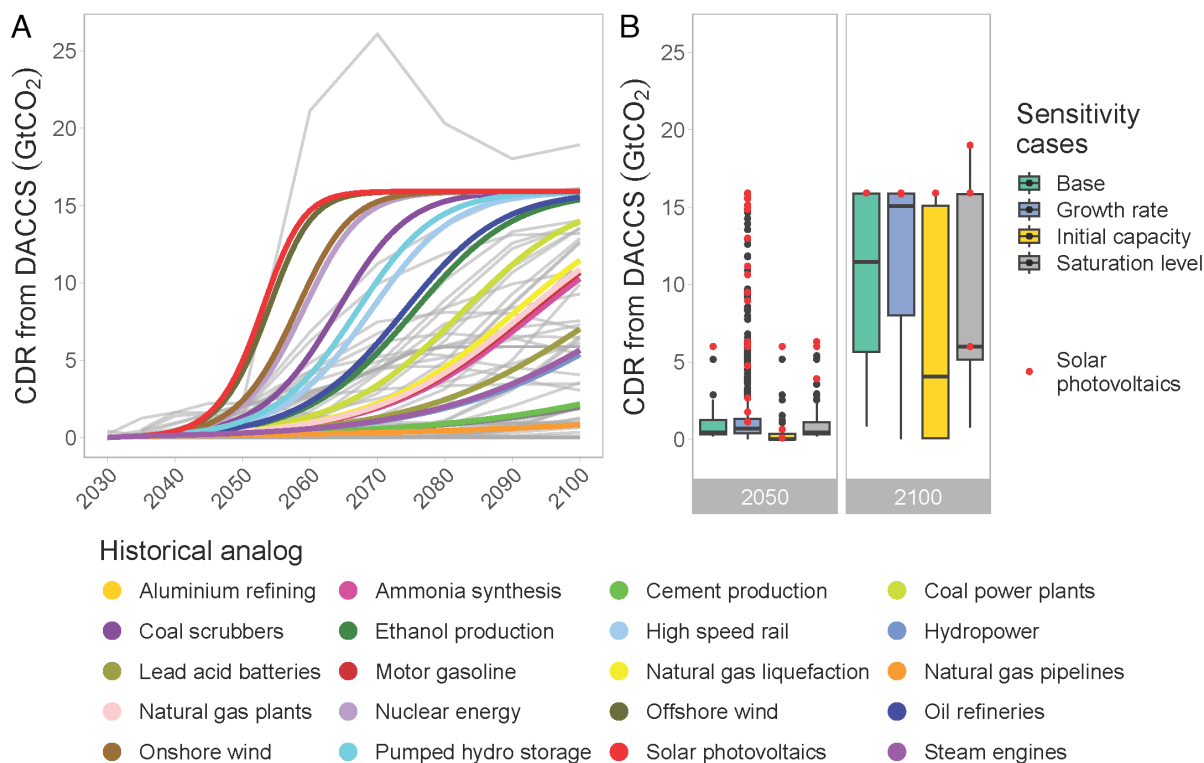
Our analysis includes three measures of early DAC adoption in increasing order of certainty that the projected capacity will be realized: capacity announcements by start-ups, capacity under construction, and current operating capacity (Fig. 2). We also note the importance of a broader set of innovation metrics for assessing progress in novel technologies such as DAC, including public grants, private investment, and government and corporate purchase agreements (*SI Appendix, section E*). To model the DACCS adoption feasibility space, we use announcements of future capacity from

start-ups as an indicator of near-term adoption potential. This metric may overestimate adoption (if realized CDR falls short of plans) or underestimate it (if CDR capacity grows rapidly as technologies derisk or new companies enter the market). We explore this sensitivity by modeling future DACCS capacity using current operating capacity and capacity under construction (Fig. 3B).

**Feasible Growth Pathways.** Projections of technology adoption, including technology growth rates over time, depend on the choice of model. Technology growth is often described using an s-curve (*SI Appendix, Note S3*), where adoption is initially low and



**Fig. 2.** Inputs to our DACCS projections using the logistic substitution model. We show (A) DACCS capacity indicators, (B) logistic growth rates from historical analogs, and (C) market for DACCS in Global Change Analysis Model (GCAM) compared to the market for DACCS and all CDR across the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6) scenarios (48). (Note that 69% of AR6 scenarios do not include DACCS as a technology option.) For our baseline projections, we use company targets in 2035 as the start of the logistic growth curve.



**Fig. 3.** Projected adoption feasibility space for DACCS using historical analogs. We show (A) baseline annual projected CO<sub>2</sub> removals from DACCS over time compared to scenarios in the IPCC AR6 database and (B) sensitivity in annual removals in 2050 and 2100 to uncertainty in the three parameters in the logistic substitution model: growth rate ( $k$ ), initial capacity ( $C_0$ ), and saturation level ( $L$ ), with solar photovoltaics shown separately in red. We use the baseline projections in A to model different DACCS adoption pathways in GCAM. See *SI Appendix, section F* for further discussion of our sensitivity analysis; box and whisker plots in B show the full range of values explored in this analysis.

grows until it reaches a saturation point (49, 50). Following this approach, we fit logistic growth curves to our historical technology analogs (see *SI Appendix, section F* for a comparison with another function form, the Gompertz curve). These curves represent cumulative technology capacity or production ( $C$ ) over time  $t$  using three parameters: the speed of adoption ( $k$ ), the inflection year ( $t_0$ ), and maturity/saturation level ( $L$ ):

$$C(t) = \frac{L}{1 + e^{-k(t-t_0)}}. \quad [1]$$

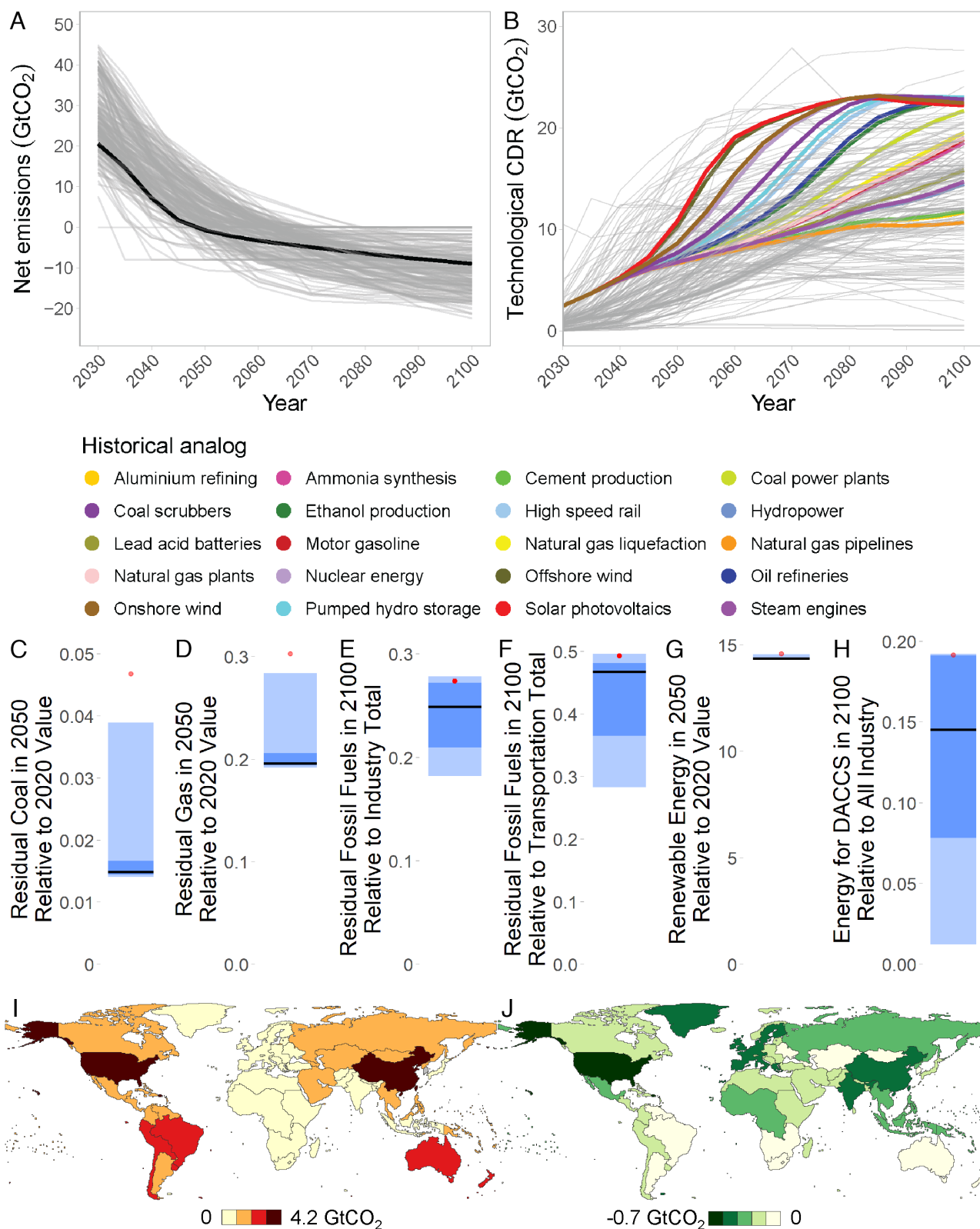
To model future growth pathways for novel technologies, we use the logistic substitution model, which uses the differential form of Eq. 1 to project future capacity using three parameters: 1) an initial capacity level (estimated using our early adoption indicators), 2) a growth rate (estimated using logistic growth rates for our historical technology analogs), and 3) a market pull or saturation level (estimated using the unconstrained demand for DACCS in GCAM in 2100, see *Materials and Methods*).

Projected adoption varies substantially across analogs, suggesting a large feasibility space for DACCS (Fig. 3). Estimated annual removals from DACCS in 2050 pass the 1 GtCO<sub>2</sub> threshold with five analogs (coal scrubbers, nuclear power, offshore wind, onshore wind, and pumped hydro storage) and solar PV, but all other analogs fall short of this target. Removals reach saturation (at the market demand of 15.9 GtCO<sub>2</sub>) in 2100 with these same analogs (except pumped hydro storage). The time when this level is reached varies across analogs, from 2074 for offshore wind (and 2072 for solar PV) to 2098 for coal scrubbers. The natural gas pipeline analog fails to reach 1 GtCO<sub>2</sub> by 2100. A lower initial capacity, such as operational and under construction DACCS, results in less than 1 GtCO<sub>2</sub> DACCS in 2050 across all analogs

and less than 1 GtCO<sub>2</sub> in 2100 for the ten analogs with growth rates less than 9.3%.

**System-wide Effects.** We use GCAM to estimate how realizing different future DACCS adoption pathways within the feasibility space would affect energy transitions across sectors and over time under a climate policy. We set an emissions constraint on global CO<sub>2</sub> emissions to limit the end-of-century temperature increase to 1.5 °C with low overshoot (51). While net emissions are constrained (Fig. 4A), total technological CDR (from DACCS and BECCS) varies across scenarios (Fig. 4B), leading to substantial differences across a variety of metrics in electricity, transportation, and industry (Fig. 4 C–H). Specifically, higher growth rates for DACCS reduce pressure for emissions in other sectors to decrease to satisfy the emissions constraint. Total energy-related emissions (excluding negative contributions from CO<sub>2</sub> sequestration) follow a similar pathway across scenarios through 2045 and then begin to diverge, with emissions varying by up to 8% in 2050 and 50% by 2075. We also see an increase in the use of BECCS in scenarios with low DACCS; however, only 23% of the difference in DACCS between our highest and lowest scenarios is made up by BECCS.

For all scenarios, massive decarbonization across electricity, transportation, and industry is needed to meet a net-zero (and net-negative) emissions target. However, differences across scenarios also emerge. We explore six metrics to quantify key differences, beginning with a focus on timelines for fossil fuel phase-down and renewable energy deployment in the electricity sector by 2050. While we observe a near-complete phaseout of coal-fired electricity across scenarios, residual generation varies slightly (from 1% of the 2020 value for the natural gas pipelines analog to 4% for offshore wind). We observe a larger range in phasedown of unabated (i.e., without CCS) electricity generation from natural gas,



**Fig. 4.** Outcomes of different DACCS pathways in the adoption feasibility space. We show global net CO<sub>2</sub> emissions (A) and CDR from DACCS and BECCS (B), compared to a range of scenarios from the AR6 database. Note CDR is on the high end of AR6 but in line with recent studies (28). We also show additional impacts of DACCS adoption on coal phaseout (C) and gas phaseout (D) in the electricity sector, residual fossil fuel consumption (without CCS) in transportation (E) and industry in 2100 (F), renewable electricity generation in 2050 (G), and energy for DACCS in 2100 (H). Black lines show the median value, and darker shading indicates the 25th to 75th percentile values. Results for the solar PV analog are shown with red dots. We also show the difference in the regional distribution of DACCS (I) and BECCS (J) in 2100 under a high DACCS scenario (offshore wind analog) and low DACCS scenario (gas pipelines analog). See *SI Appendix, Figs. S8–S19* for additional output variables.

from 19 to 28% of the 2020 value. For both metrics, there is a substantial increase with the higher growth analogs. Differences in generation from variable renewable energy sources are smaller

(approximately 1% difference) due to compensating effects in industry electricity demand (including electricity for DACCS) and a smaller change in transportation electrification.

Timelines for fossil fuel phasedown and renewable energy deployment are important for climate policy as they can inform decisions to invest (or not) in new fossil fuel infrastructure, plans to manage stranded assets (52), and intermediate targets for technology deployment. As the model time horizon extends to the end of the century, the portfolio of mitigation technologies (including but not limited to DACCS) is increasingly uncertain. Recognizing these limitations, we focus on residual emissions as a metric for understanding the level of innovation required in so-called hard-to-abate sectors (53). Residual unabated fossil fuel use in 2100 varies from 28 to 49% in transportation and 18 to 28% in industry as a fraction of total energy inputs (again comparing the natural gas pipeline and offshore wind analogs). Residual emissions also have important temporal dynamics, rebounding after mid-century and then declining by 2100, with higher rebounds seen for fast-growing analogs that saturate earlier (*SI Appendix, Figs. S6–S11*). Additionally, up to 19% of total final industrial energy use in 2100 goes to DACCS.

Different DACCS adoption pathways also influence the regional distribution of CO<sub>2</sub> emissions and removals under a climate policy (Fig. 4 *I* and *J* and *SI Appendix, Figs. S20 and S21*). Because our scenarios consider a global emissions policy, constraints on adoption pathways for key technologies such as DACCS can influence where mitigation is most cost-effective and thus the global distribution of mitigation burdens from both low-carbon technologies and CDR. The largest CO<sub>2</sub> removals from DACCS occur in the United States, China, and Brazil (up to 4.1, 3.2, and 2.4 GtCO<sub>2</sub> per year, respectively). Higher DACCS growth alleviates pressure for CDR from BECCS, with potential benefits in terms of land and water demand and thus food scarcity (11). The greatest reductions in CDR from BECCS are again seen in the United States and China but also India. Due to the need to reduce fossil fuel use, regions with high CDR or dependence on fossil fuels (e.g., oil producing regions) experience the largest differences in emissions across the DACCS feasibility space.

## Discussion

Many technologies that will be important for climate change mitigation are in the early stages of development, and their future growth is highly uncertain. We develop an approach to model the adoption feasibility space for novel technologies and apply it to DACCS. Our approach combines empirical data on historical technology analogs, early adoption indicators, and a detailed process-based IAM. We find that there is substantial variation in the projected role of DACCS in climate policy across historical analogs, with some projections reaching many GtCO<sub>2</sub> in removals by the end of the century and others with very low adoption. The contribution of DACCS to meeting a 2050 net-zero target is relatively smaller but nevertheless implies different requirements for the phasedown of natural gas (and to a much lesser extent coal). By the end of the century, higher DACCS adoption with several high-growth analogs could reduce pressure on to reduce residual fossil fuel emissions. The largest differences across scenarios are seen in regions with high CDR from DACCS and/or BECCS and those with high dependence on fossil fuels.

Our motivation for developing this approach to model novel technology adoption derives from its potential to overcome the deficiencies of alternative approaches. Allowing modeled technologies to grow to satisfy emissions targets may exceed real-world limits. However, imposing fixed constraints on growth rates lacks an empirical basis (31). Eliciting expert predictions about the future risks codifying ignorance (54). We thus use the one laboratory large

enough to characterize innovation systems: historical technologies (55). One limitation is that this approach can be sensitive to the data inputs (Fig. 3*B* and *SI Appendix, section F*). The sensitivity of long-term adoption to near-term values highlights the importance of early policy support. Additionally, methods using analogs have been critiqued for removing technologies from their historical context (56). While we cannot fully address this critique, the approach we develop aims to identify strategic similarities between analogs and innovations and uses a range of analogs, making it less sensitive to artifacts of a single technology (45). Further, the contexts in which analogs grew can point to important ways to support DACCS.

The approach we present here could be extended to represent other novel technologies in systems models. Under stringent mitigation targets, pathways for CDR can be less sensitive to assumptions about costs than about growth constraints. This is because, absent other approaches to reduce emissions in hard-to-abate sectors, substantial CO<sub>2</sub> removal is needed at almost any cost (28). However, for other technologies, variables such as cost and performance may be larger drivers of model outcomes, and historical analogs may be used to project these variables. Future work could also examine spatial patterns in the adoption of historical technologies to model technology diffusion dynamics across regions. For these applications, it will be critical to understand the limits of prediction using historical analogs, including the ability to infer systematic patterns in early and late adopters (23) and distinguish between saturation and temporary slow-downs in technology adoption (57). Ultimately, these model improvements should be made with an eye toward identifying robust mitigation pathways given technology uncertainties and opportunities to learn over time (58).

## Materials and Methods

**Integrated Assessment Model.** We use the GCAM version 6.0 in this research (available at <https://github.com/JGCRI/gcam-core/releases>). We run GCAM through the University of Wisconsin-Madison Center for High Throughput Computing (CHTC) (59). GCAM is an open-source IAM developed and maintained by Pacific Northwest National Laboratory and is widely used to design scenarios to inform climate policy. GCAM is a dynamic-recursive model (i.e., agents make decisions sequentially in each period based only on information in that period) and represents relationships among five global systems at different spatial scales: energy, socioeconomic, land and agriculture, water, and climate. Market equilibrium is the operating principle for GCAM. For each five-year time step, GCAM solves for a set of prices such that supply equals demand for all markets and sectors across all regions (32 geopolitical regions, 384 land-water regions). GCAM projects outcomes through 2100 at 5-y intervals, including energy supply and demand, regional prices (e.g., electricity, agricultural commodities, etc.), emissions of greenhouse gases and short-lived species, and climate outcomes.

GCAM is a technology-rich model with a detailed regional representation of energy technologies and end use in buildings, transportation, and industry as well as agriculture and land use. Technological innovation is represented by changes in parameters such as cost (energy system) or profit (land and agriculture) and performance (e.g., efficiency of energy storage or fertilizer requirements for agricultural products). GCAM uses a logit formulation to represent local variation in preferences and other factors that enable technologies that are theoretically inferior in cost or profit alone to capture some market share. GCAM also represents changes in technology availability (for example, due to breakthrough innovations) using share weights. GCAM includes a variety of parameters that limit land conversion for biomass (including a price on CO<sub>2</sub> emissions from land use change). However, there are tradeoffs across CDR methods, and we do observe higher levels of BECCS in low DACCS scenarios. Specifically, GtCO<sub>2</sub> removal from BECCS totals 6.3 GtCO<sub>2</sub> in 2100 in our highest DACCS growth scenario (offshore wind analog) and 9.8 GtCO<sub>2</sub> in our lowest DACCS growth scenario (natural gas pipelines).

We model a 1.5 °C climate policy in GCAM by applying a global constraint to net CO<sub>2</sub> emissions. The constraint is applied to each five-year time step beginning in 2025 and declines linearly from 8,018 MtC to 480 MtC in 2045 then –602 MtC in 2050. After 2050, net CO<sub>2</sub> emissions reach –2206 MtC in 2100. These scenarios reach a peak temperature rise of 1.8 °C in 2055 before declining to 1.5 °C in 2100. This emissions pathway is in the middle of the range of AR6 scenarios (Fig. 4A) (28). We apply constraints on DACCS by specifying adoption at each five-year time step beginning in 2025. GCAM selects the proportion of DACCS provided by three processes: low temperature (S-DAC) with an electric heat pump and high temperature (L-DAC) with electricity or natural gas. The proportion of DACCS from S-DAC ranges from 21 to 23% in 2050 to 32% in 2100 across scenarios. Since our scenarios only constrain CO<sub>2</sub>, emissions of other climate-warming species vary slightly with minimal effects on temperature. (See *SI Appendix, Note S6* for a discussion of how GCAM compares to other IAMs.)

**Historical Technology Analogs.** We apply the SHARD method to identify historical analogs for DACCS (45) (*SI Appendix, Note S4*). The SHARD method consists of three steps. We briefly describe each step here and provide a full discussion, including justifications for the selection of our specific analogs, in *SI Appendix, section B*:

1. Clarify objectives. The objective of this study is to quantify the feasibility space for future DACCS adoption and model the effects of different DACCS pathways within this feasibility space under a 1.5 °C climate policy. Given this objective, we focus on identifying multiple analogs that span the feasibility space and capture different features of DACCS technologies and their use context.
2. Develop a long list. For the long list of technologies we consider as analogs for DACCS, we use the Historical Adoption of TeCHnology (HATCH) database, which contains capacity growth over time for 148 technologies spanning 10 categories: agriculture, chemicals, digitalization, energy supply, household appliances, infrastructure, materials, space and defense, energy storage, and miscellaneous (60). After applying various data cleaning steps (*SI Appendix, section B*), we arrive at a long list of 61 technologies.
3. Select technologies. We select technology analogs to strategically represent different features of the DACCS process, including an initial filtering process based on broad technology and infrastructure and user and consumption characteristics followed by a more detailed examination of technologies across other dimensions of the SHARD method (*SI Appendix, Note S4*).

We source technology data from HATCH with some modifications and additions (described in *SI Appendix, section B*). We model scenarios where DACCS adoption is constrained to grow at rates observed for different analogs.

**Early Capacity Data.** We evaluate company announcements to generate empirical estimates of early DAC capacity. Data on announcements are sourced from recent reports and company websites (*SI Appendix, section C and Table S1*). DAC companies forecast a total annual capture of 113 Mt CO<sub>2</sub> in 2035, of which Carbon Engineering accounts for 100 Mt CO<sub>2</sub> plants. The remaining largest company is CarbonCapture, whose Project Bison uses a modular DAC technology with plans to reach 5 Mt CO<sub>2</sub> annual capacity by 2030. Companies with plans for construction but without start-up dates are not included, such as CarbonCollect's mechanical trees. We use three different initial capacities to project future DACCS capacity: operating capacity as of 2023, capacity under construction (which is scheduled to be completed by 2025), and construction targets for capacity to come online by 2035. Our baseline scenarios project DACCS from company forecasts. Our lower-bound scenarios project future DACCS from currently operating

plants (0.046 MtCO<sub>2</sub>), and we also consider a scenario using capacity under construction (0.588 MtCO<sub>2</sub>).

**Projecting Technology Adoption.** We project technology adoption using the logistic substitution model, which uses the differential form of the logistic growth model (Eq. 1). The discrete form of the logistic substitution model can be used to iteratively calculate capacity  $C$  at time  $t$  based on an initial capacity value (*SI Appendix, Note S3 and section D*):

$$C_t + \frac{dC_t}{dt} = C_{t+1} = C_t \left( 1 + k \left( 1 - \frac{C_t}{L} \right) \right), \quad [2]$$

where  $k$  is the growth rate (taken from a historical analog) and  $L$  is the saturation level or market pull. We estimate market pull as the total market for DACCS in GCAM in 2100 under our climate policy. This demand reflects technical features of DACCS (9) and competition with other climate solutions but not additional contextual factors that may enable or constrain growth. DACCS reaches 15.9 GtCO<sub>2</sub> in annual removals in 2100 in this scenario, and CDR from BECCS and DACCS combined reaches 22.5 GtCO<sub>2</sub>, which is on the high end of the range of AR6 scenarios (Fig. 4B and *SI Appendix, Note S6*). This high market demand for CDR makes GCAM an attractive model for our study because it can simulate the full feasibility space for DACCS that we calculate using a range of high- and low-growth analogs.

**Data, Materials, and Software Availability.** All study data are included in the article and/or *SI Appendix*. The code for this analysis can be found at <https://doi.org/10.5281/zenodo.10970811> (61).

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